

Lab 8: Unsupervised Learning Mini-Project

Karla Dominguez Rivera

PART 1 Preparing the data

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
```

```
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	

842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
```

```
diagnosis <- factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

There is 569 observations.

```
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

212 observations have malignant diagnostics.

```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

****Q3.**How many variables/features in the data are suffixed with `_mean`?

There is 10 variable features in the data that are suffixed with `_mean`.

```
sum(grepl("_mean", names(wisc.df)))
```

```
[1] 10
```

PART 3. Principal Component Analysis

```
# Check column means and standard deviations  
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean

7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
# Perform PCA on wisc.data by completing the following code
```

```
wisc.pr <- prcomp(wisc.data, scale. = TRUE)
```

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

The first principal component (PC1) captures about 44.27% of the total variance in the data.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```
pr.var <- wisc.pr$sdev^2
pve <- pr.var / sum(pr.var)
cumulative_pve <- cumsum(pve)
which(cumulative_pve >= 0.70)[1]
```

```
[1] 3
```

You need 3 principal components to explain at least 70% of the variance.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

```
pr.var <- wisc.pr$sdev^2
pve <- pr.var / sum(pr.var)
cumulative_pve <- cumsum(pve)
which(cumulative_pve >= 0.90)[1]
```

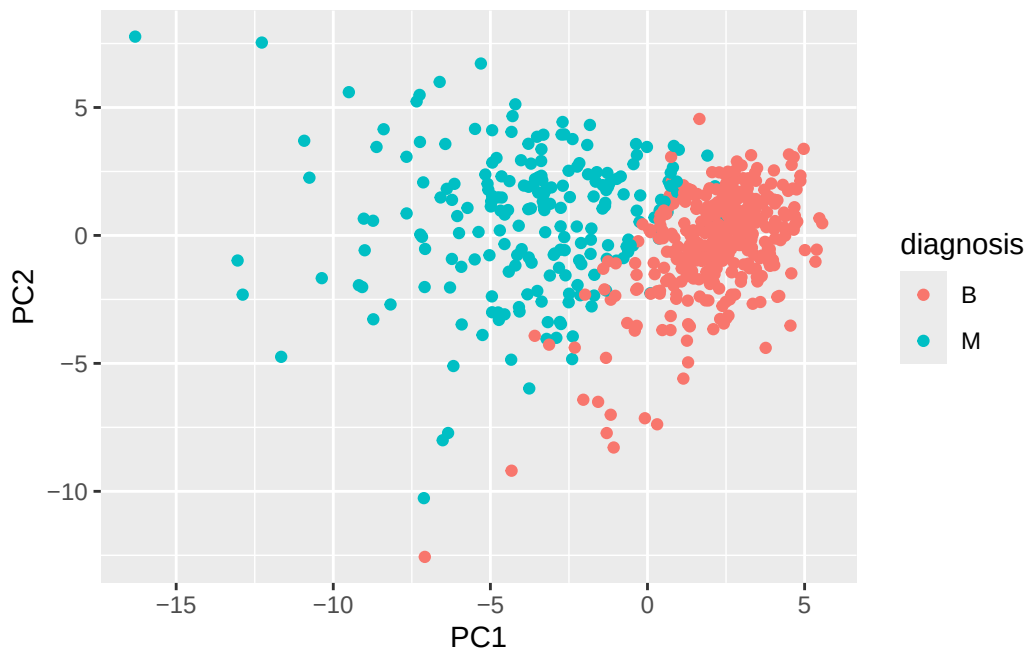
```
[1] 7
```

You need 7 principal components to explain at least 90% of the variance.

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The biplot is very cluttered and difficult to interpret with many data points and overlapping each other. It is hard to clearly see patterns or groupings in the data making the plot not very useful for understanding the PCA results.

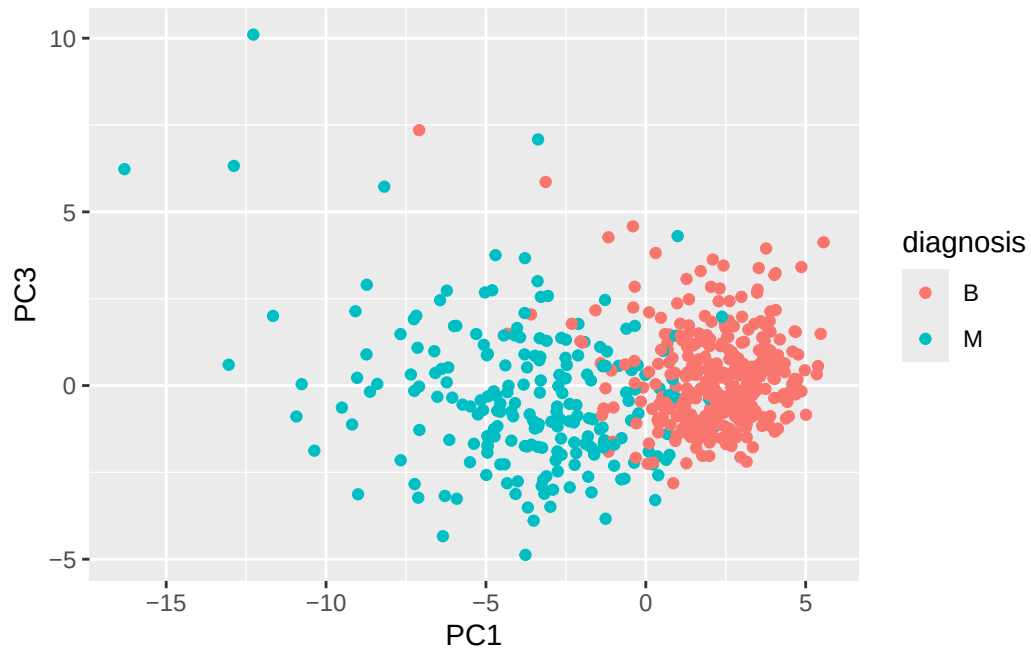
```
biplot(wisc.pr)
```

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The plot of PC1 vs PC3 still shows separation between benign and malignant cases, but it is not as clear as PC1 vs PC2. PC1 appears to be the strongest component for distinguishing the groups. PC3 adds some variation, but it does not separate the classes as well as PC2 does in combination with PC1.

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col = diagnosis) +  
  geom_point()
```

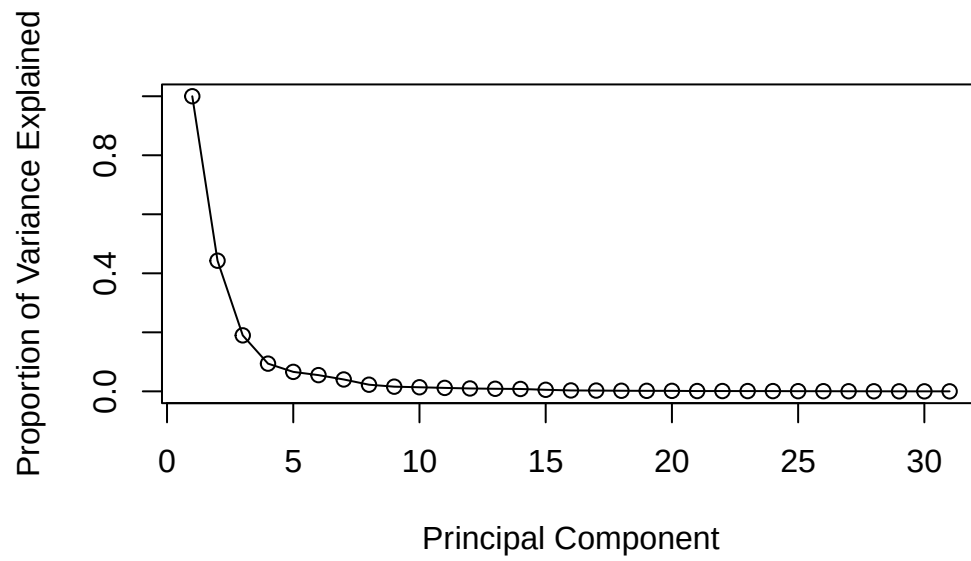


```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

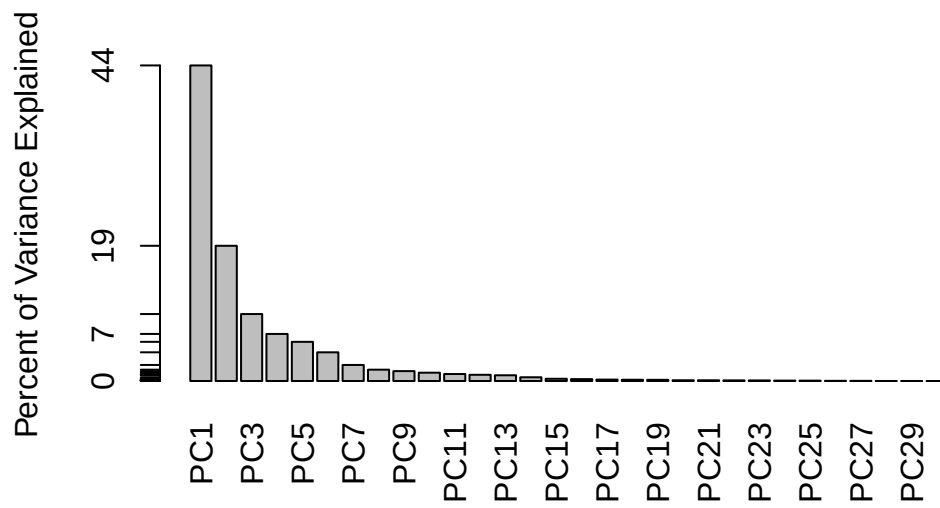
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

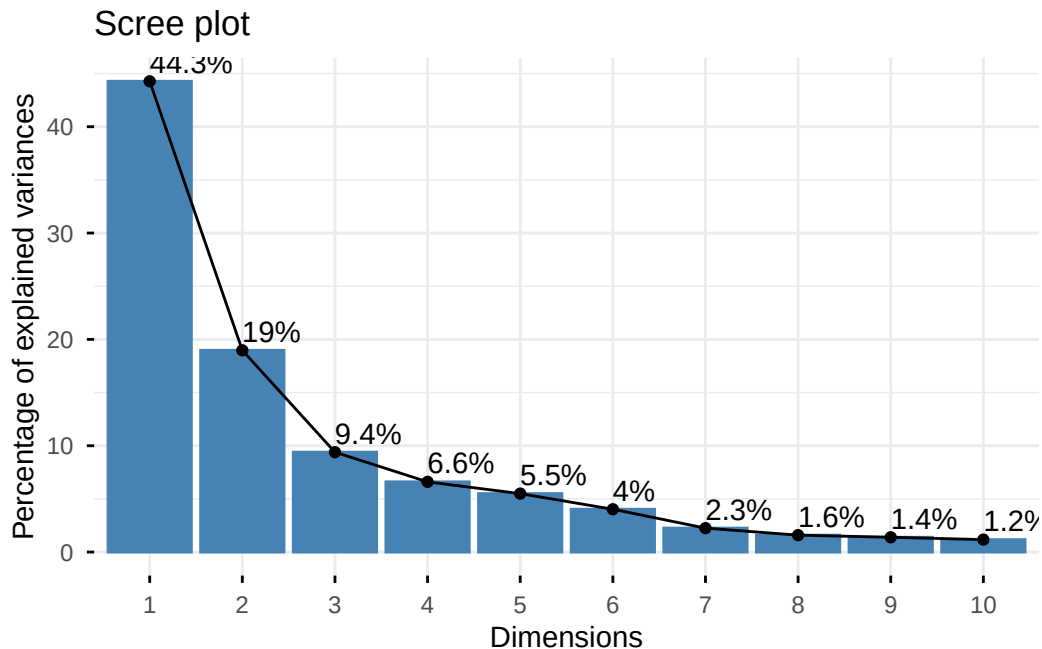


```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product-guide-to-principal-component-methods-in-r/>

```
fviz_eig(wisc.pr, addlabels = TRUE)
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The loading value for `concave.points_mean` on PC1 is large and positive around 0.26. Yes, there are a few features with even larger contributions to PC1.

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

PART 4: Hierarchical clustering

```
data.scaled <- scale(wisc.data)
```

```
data.dist <- dist(data.scaled)
```

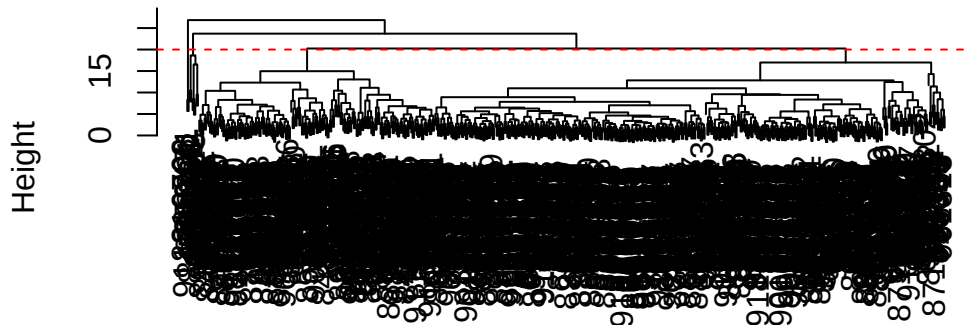
```
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

The height where the clustering produces 4 clusters is approximately 20.

```
plot(wisc.hclust)
abline(h = 20, col = "red", lty = 2)
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

Yes, you can try different numbers of clusters. Often, 2 clusters already gives a very strong separation because the data naturally splits into benign vs malignant

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
```

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

the method ward.D2 gives the best results because it creates the most compact and well-separated clusters, which most closely match the true benign vs malignant labels in the dataset.

```
wic.hclust.complete <- hclust(data.dist, method= "complete")
wic.hclust.average <- hclust(data.dist, method= "average")
wic.hclust.single <- hclust(data.dist, method= "single")
wic.hclust.ward <- hclust(data.dist, method= "ward")
```

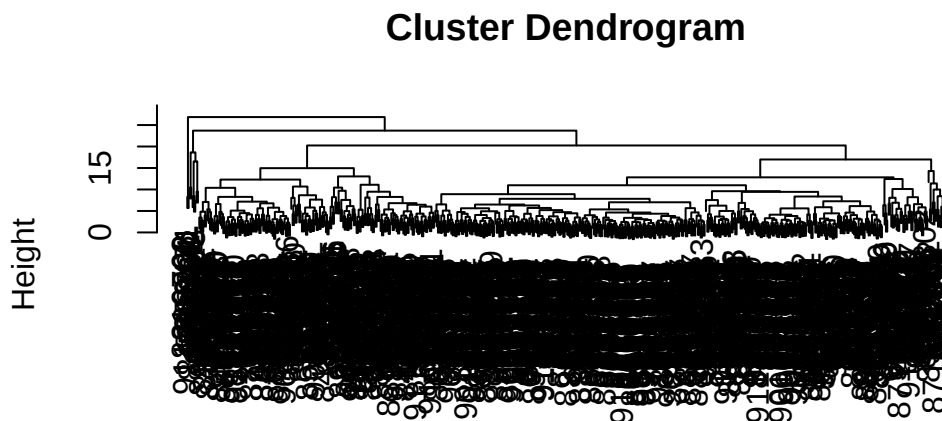
The "ward" method has been renamed to "ward.D"; note new "ward.D2"

```
View(wisc.df)
```

PART 5: Combining methods

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
```

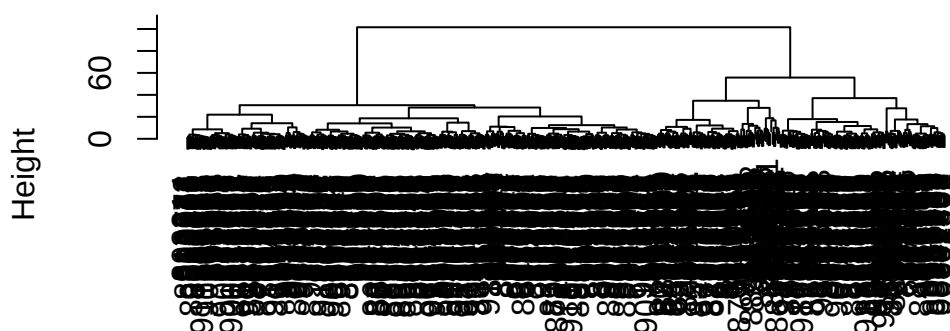
```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
plot(wisc.hclust)
```



data.dist
hclust (*, "complete")

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
plot(wisc.pr.hclust)
```

Cluster Dendrogram



```
dist(wisc.pr$x[, 1:7])
hclust (*, "ward.D2")
```

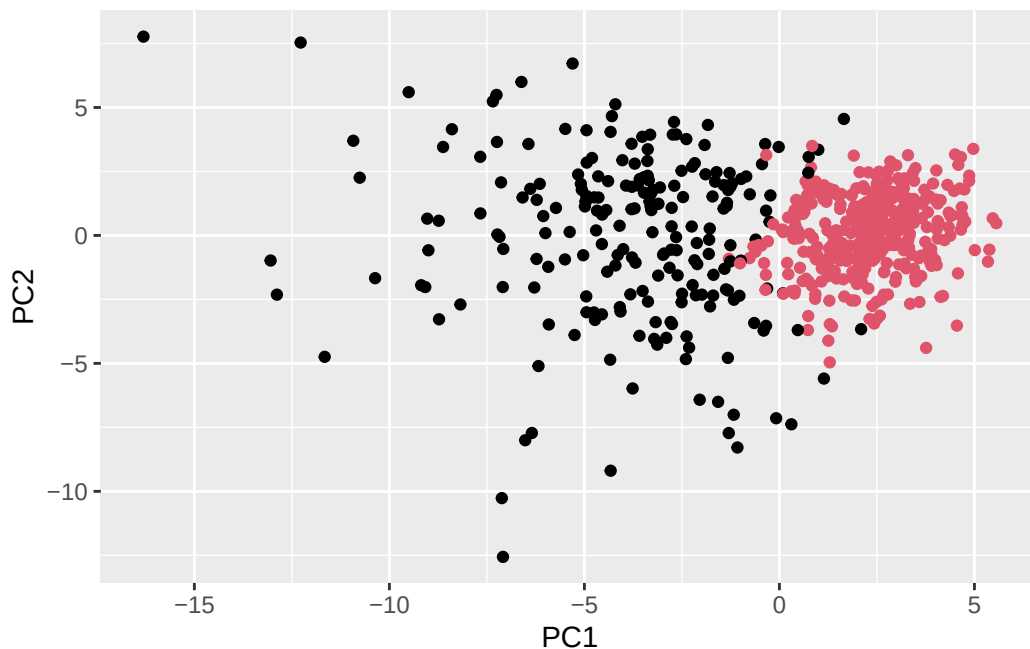
```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
216 353
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1  28 188
  2 329  24
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



optional :

```
library(rgl)
plot3d(wisc.pr$x[,1:3], xlab="PC 1", ylab="PC 2", zlab="PC 3", cex=1.5, size=1,
       type="s", col=grps)
```

```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
```

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
```

Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24
```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

The hierarchical clustering model without PCA does not separate the diagnoses very well. Several clusters contain a mixture of benign and malignant cases instead of clean separation.

```
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B   M
1      12 165
2       2   5
3     343  40
4       0   2

```

PART 6 : Sensitivity/Specificity

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

The PCA-based hierarchical clustering model gives the best performance overall. It has the highest sensitivity because it correctly identifies most malignant cases.

***PART 7 :Prediction**

```

#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc

```

```

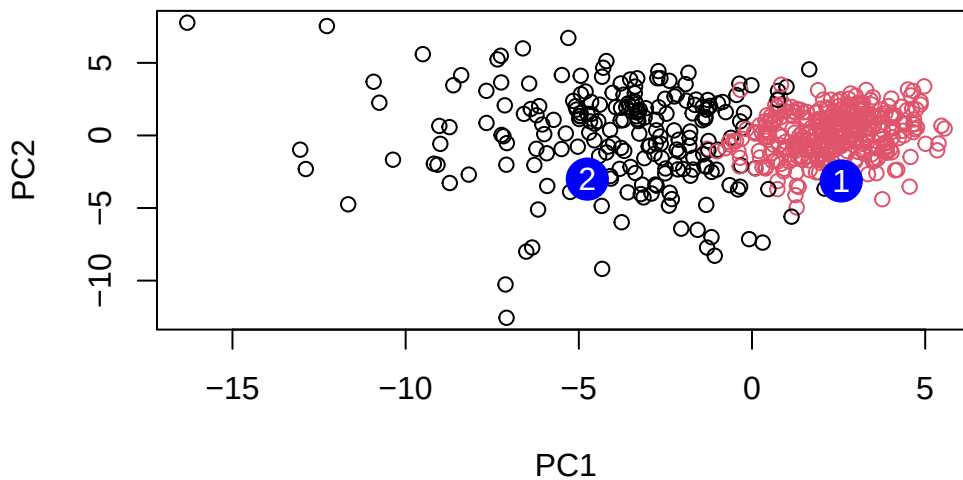
      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10      PC11      PC12      PC13      PC14
[1,] -0.2307350  0.1029569 -0.9272861  0.3411457  0.375921  0.1610764  1.187882
[2,] -0.3307423  0.5281896 -0.4855301  0.7173233 -1.185917  0.5893856  0.303029
      PC15      PC16      PC17      PC18      PC19      PC20
[1,]  0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,]  0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500

```


	PC21	PC22	PC23	PC24	PC25	PC26
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238

	PC27	PC28	PC29	PC30
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Prioritize Patient 1 for follow-up because Patient 1's PCA scores place them closer to the cluster associated with malignant tumors, which suggests a higher risk. Patient 2 appears to fall in the region more associated with benign cases, so they are lower priority.